

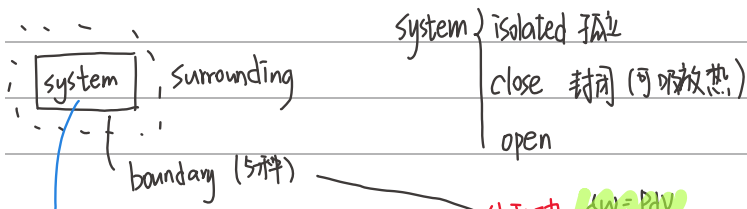


1. 热力学 { 宏观: Macroscopic (宏观)
微观: Statistical

2. 相变

chapter 1

1-1

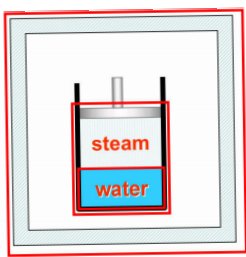


energy 交换 (热) → Heat
matter 交换

其他形式为 work } 体元功 $\delta W = PdV$
其他

- Mechanical work $\delta W = f_c dl$
- Electric work $\delta W = Edq$
- Surface work $\delta W = \gamma dA_s$

adiabatic wall



Isolated system
Steam+water is a closed system
Water is a open system

Thermal energy 内能
Heat 热量 (传递)

- **Adiabatic:** No thermal energy can pass through.
- **Diathermal:** Thermal energy can pass through.
- **Permeable:** Matter can pass through.
- **Impermeable:** Matter cannot pass through.
- **Semipermeable:** Some components are able to pass through, while others are not.

Energy of a system } Internal energy 内能
 Potential energy 势能
 kinetic energy 动能 (微观上考虑)

热能 $\xrightarrow{\text{热机}}$ 机械能

1-2 state 状态 } 微观 Microscopic state

宏观 T, P, V 并不完全独立 Macroscopic state

体积变化 } 等温压缩 $\beta_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{\text{等}T}$

热膨胀 $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{\text{等}P}$

定值 for ideal gas $\alpha = \frac{1}{273.15}$ 理想气体 分子之间无相互碰撞
绝对零度

$F = C - P + 2$ number of coexisting phase

$\downarrow$ number of components

degree of freedom $F=1$: 线移动 $F=2$: 面移动

chapter 2:

$$\Delta U = U_B - U_A = q - w$$

$\rightarrow$ first law

$$dU = \delta q - \delta w \quad (q, w \text{ 都与路径有关})$$

heat capacity:

$$C = \frac{\delta q}{dT} \quad \left\{ \begin{array}{l} V \text{ 不变 } C_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{dU}{dT} \right)_V \quad w=0 \\ P \text{ 不变 } C_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH}{dT} \right)_P \end{array} \right.$$

$$\Delta H = \Delta U + P(\Delta V) = \Delta U + \Delta W = q$$

$$\left\{ \begin{array}{l} C_V = n C_{V,m} \\ C_P = n C_{P,m} \end{array} \right.$$

$$\begin{cases} C_p = \left(\frac{dq}{dT}\right)_p = \left(\frac{dH}{dT}\right)_p = \left(\frac{d(u+Pv)}{dT}\right)_p = \left(\frac{\partial u}{\partial T}\right)_p + P\left(\frac{\partial v}{\partial T}\right)_p \\ C_v = \left(\frac{dq}{dT}\right)_v = \left(\frac{du}{dT}\right)_v \end{cases}$$

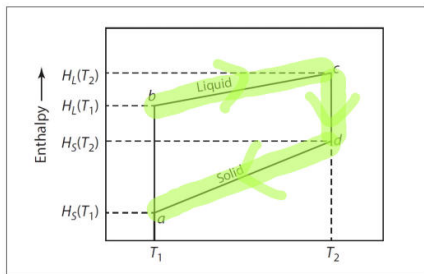
$$C_p - C_v = \left(\frac{\partial u}{\partial T}\right)_p + P\left(\frac{\partial v}{\partial T}\right)_p - \left(\frac{\partial u}{\partial T}\right)_v = P\left(\frac{\partial v}{\partial T}\right)_p$$

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv \quad u(v, T)$$

$$\downarrow$$

等时 $du = \left(\frac{\partial u}{\partial v}\right)_T dv = p_{inter} \cdot dv$ 内压

$$\begin{cases} dH = C_p dT \\ du = C_v dT \end{cases} \quad \text{理想气体任何过程}$$



$$Hb \rightarrow c = \int_{T_1}^{T_2} C_p^L dT = C_p^L (T_2 - T_1)$$

$$Hd \rightarrow a = \int_{T_2}^{T_1} C_p^S dT$$

都要先算 n . $n = \frac{PV}{RT}$

绝热条件:

$$\begin{cases} du = C_v dT \\ \delta w = PdV \end{cases} \Rightarrow C_v dT = -PdV, \quad P_1 V_1^r = P_2 V_2^r = \text{constant}$$

reversible 可用

$$C_p = \frac{5}{2}R, \quad C_v = \frac{3}{2}R, \quad r = \frac{5}{3}$$

$$W = -\Delta U = -\int_{T_a}^{T_b} C_v dT = -C_v (T_b - T_a) \quad \text{也可用 } W = \int P dV = P V^{\gamma} \times \int \frac{1}{V^{\gamma}} dV$$

$$\Delta H = \int C_p dT$$

等温条件:

$$dU = 0 \quad PV = \text{constant} \quad \text{reversible 可用}$$

$$W = q = \int P dV = n \int \frac{RT}{V} dV = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$

$$\Delta H = \Delta U + (P_1 V_1 - P_2 V_2) = P_1 V_1 - P_2 V_2 = nRT_1 - nRT_2 = 0$$

等压过程:

$$\Delta U = \int C_v dT \quad W = P(V_2 - V_1)$$

$$q = \Delta H = \int C_p dT \quad \Delta H = (U_2 + P_2 V_2) - (U_1 + P_1 V_1) = \Delta U + P(V_2 - V_1) = \Delta U + dW = q$$

等容过程 Isochoric

$$\Delta U = \int C_v dT$$

$$\Delta H = \int C_p dT, \quad q = \Delta U, \quad W = 0$$

chapter 3 (精)

natural / spontaneous: non-equilibrium $\rightarrow$ equilibrium

$\rightarrow$ 自发且不可逆

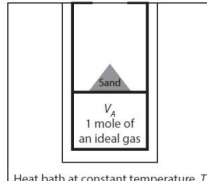
1. $\Delta S = \frac{q_{\text{reversible}}}{T}$ (温度本身) 和 $C = \frac{q}{\Delta T}$ 区分

$\Delta S = \frac{q_{\text{re}}}{T} = \frac{q}{T} + \Delta S_{\text{ire}}$
 $\rightarrow$ 由于反应不可逆而引起的熵.

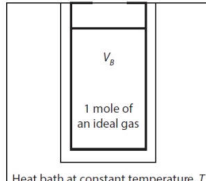
a reversible process: 一直平衡态.

a quasi-static process: 在无限小力的影响下.

eg.



(a)



(b)

每次拿取无限小的沙砾
等温膨胀

$\Delta U = 0$
 $q = w = RT \ln \frac{V_B}{V_A}$

$\Delta S_{\text{gas}} = \frac{q_{\text{rev}}}{T} = \frac{w_{\text{rev}}}{T} = R \ln \frac{V_B}{V_A} > 0$

由于整体平衡:

$\Delta S_{\text{heat reservoir (热源)}} = \frac{-q}{T} = -\Delta S_{\text{gas}} = R \ln \frac{V_A}{V_B} < 0$

$\Delta S_{\text{total}} = 0$

irreversible: (free expansion. T不变, $P=0$)

由于 S 是状态量, 气体与 reversible 的初末状态一致. (T, V 一致) 则 ΔS 同上

$\Delta S_{\text{gas}} = R \ln \frac{V_B}{V_A}$

但因为 $\Delta U = 0$ 且 $P = 0 \therefore q = 0$ 没有热量转移

$\Delta S_{\text{heat reservoir}} = 0$ (环境无限大, 视为 reversible)

$\therefore \Delta S_{\text{total}} = \Delta S_{\text{gas}} + \Delta S_{\text{热源}} = \Delta S_{\text{gas}} = R \ln \frac{V_B}{V_A} > 0$

∴ 可逆 $\Rightarrow$ 熵增.

※ 等温可逆过程中: $W_{\max} = q_{\text{rev}}$ W 做的功最多, q 的转移最多

∴ 初末状态相同时 $0 \leq W \leq W_{\max}$, $0 \leq q \leq q_{\text{rev}}$, $0 \leq \Delta S_{\text{total}} \leq \frac{q_{\text{rev}}}{T}$

reversible 上限



free expansion 下限

※ 蒸发过程: 可逆: $W_{\max} = P_{\text{external}} V$

不可逆: $W = (P_{\text{external}} - \Delta p) V$

degraded work

$$W_{\max} - W = q_{\text{rev}} - q$$

$$\Delta S_{\text{heat reservoir}} = -\frac{q}{T}$$

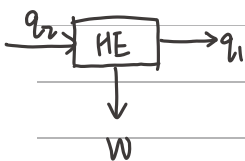
$$\Delta S_{\text{water + vapor}} = \frac{q}{T} + \frac{q_{\text{rev}} - q}{T} = \frac{q_{\text{rev}}}{T}$$

$$\Delta S_{\text{total}} = \frac{q_{\text{rev}} - q}{T} \quad \Delta S_{\text{irr}}$$

绝热 curve 上每 - state 都等熵. (isentropic) $q_{\text{rev}} = 0$. $\Delta S = 0$

1. Entropy is a thermodynamic state variable (function).
2. Entropy is not created when a system undergoes a reversible process; entropy is transferred from one part of the system/surroundings to another part.
3. The total entropy of the universe increases when an irreversible process occurs.
4. For all processes, we can write $\Delta S_{\text{system}} = q/T + S_{\text{irr}}$ and $q \leq q_{\text{rev}}$.
5. For all processes, the entropy of the universe increases or stays the same. The total entropy of the universe never decreases.

二. 热机 Heat engine



$$\Delta U = q_2 - q_1 - W$$

卡诺循环:

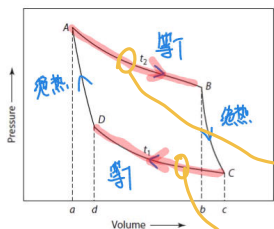


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ 全为 reversible

循环从环境吸 q_2 热量到热机中

做的热量 q_1

$$\eta = \frac{W}{q_2} = \frac{q_2 - q_1}{q_2} = 1 - \frac{q_1}{q_2}$$

卡诺循环的 $\Delta S = 0$.

$$\frac{q_2}{T_2} = -\frac{q_1}{T_1}$$

卡诺循环的 η 只与上下温度有关

$$\frac{q_1}{q_2} = f(T_1, T_2)$$

$$\frac{q_1}{q_2} \times \frac{q_2}{q_2} = \frac{f(T_1, T_2)}{f(T_1, T_2)} = \frac{q_1}{q_2} = f(T_1, T_2)$$

$$\text{令 } f(T_1, T_2) = \frac{F(T_1)}{F(T_2)}$$

$$\frac{q_1}{q_2} = \frac{F(T_1)}{F(T_2)} = \frac{T_1}{T_2}$$

$$\text{then } \frac{q_1}{q_2} = \frac{T_1}{T_2}$$

$$\eta = 1 - \frac{q_1}{q_2} = 1 - \frac{T_1}{T_2}$$

$$dS_{\text{system}} = \frac{\delta q}{T} + dS_{\text{irr}}$$

道关于 ΔS 和 ΔS_{irr} 的例题。

5 mol 单原子理想气体。绝热。50 atm, 300 K。压力突然降为 10 atm (irreversible) expansion, 且体积功 4000 J。求：

1. 最终 irreversible 温度 T_1 。若是 reversible 的温度 T_2 。

reversible: $pV^\gamma = \text{constant}$ 只有 reversible 可用。

$$p_0 V_0 = nRT_0 \Rightarrow V_0 = \frac{nRT_0}{p_0} = \frac{5 \times 0.08206 \times 300}{50} = 2.46 \text{ liters}$$

$$p_0 V_0^{\frac{5}{3}} = p_1 V_1^{\frac{5}{3}} \Rightarrow V_1 = 6.46 \text{ liters}$$

$$T_1 = \frac{p_1 V_1}{nR} = 157.5 \text{ K}$$

irreversible:

$$\Delta U = q - w, q = 0. \Delta U = -w$$

$$\Delta U = \int_{T_1}^{T_2} n C_V dT = 5 \times 1.5 \times 8.314 \times (T_2 - 300) = -4000 \Rightarrow T_2 = 233.9 \text{ K}$$

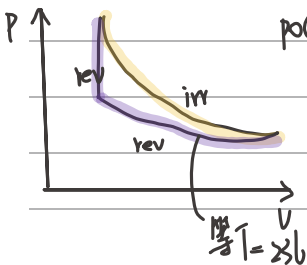
2. 计算 irreversible 过程的热量。

$$dS = \frac{\delta q}{T} + dS_{\text{irr}}$$

任意可逆过程 a 的 q 和 T 。

$$= \frac{q_{\text{rev}}}{T} \text{ 必须为 reversible 过程}$$

∴ 本题要求 $dS_{\text{irr}} \rightarrow$ 2个 reversible $= \frac{\delta q}{T} + dS_{\text{irr}}$ 其中 $\frac{\delta q}{T} = 0$ (因为绝热。)



process 1: $P: 50 \text{ atm} \rightarrow P_2$, $V: 2.46 \text{ liter}$, $T: 300 \rightarrow 236$

$$\Delta S = \frac{q}{T} = \frac{n C_V dT}{T} \stackrel{(w=0)}{=} = 5 \times 1.5 \times 8.314 \times \ln \frac{236}{300} = -15.5 \text{ J/K}$$

process 2: $(\Delta U = 0)$

$$q = w = nRT \ln \frac{V_2}{V_1}, V_1 = 2.46$$

$$V_2 = \frac{nRT}{P} = \frac{5 \times 0.08206 \times 298}{10} = 9.68$$

$$\Delta S = \frac{Q}{T} = 57.0 \text{ J/K}$$

$$\Delta S = \Sigma \Delta S = 41.5 \text{ J/K}$$

Chapter 4 S 和 Ω 的关系

entropy: degree of mixed-up-ness 混乱程度

} classical thermodynamics. 用 P, V, T 中任2个来描述1个独立热力学系统

} statistical thermodynamics atom

↓ } Isolated system: $S = S(U, V, N)$ or $U = U(S, V, N)$

closed system: $S = S(T, V, N)$

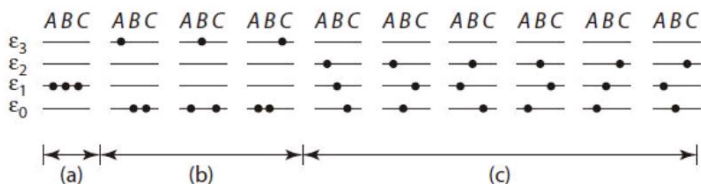
open system: $S = S(T, V, \mu)$

微观的平衡态: the most probable of all of its possible ^{宏观} 微观概率最大 $\rightarrow$ 平衡

微观 particle 中存在能级差 forbidden energy bands $\epsilon_1, \epsilon_2, \epsilon_3$ (ϵ_0 基态)

↓
ground state

假设3个粒子 ABC. $\epsilon_0 = 0$, $\epsilon_1 = u$, $\epsilon_2 = 2u$. ABC总 $U = 3u$ 则有10种分布可能:



10种 microstates, 3种 distribution

3种 distribution C 微观

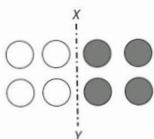
U, V, N 相同 $\rightarrow$ 宏观只有1种.

概率最大 $\rightarrow$ C 为宏观平衡态.

当粒子有无数多时 $\Omega_{\max}$ (most probable) 远远大于其他 $\therefore$ 认为 $\Omega \approx \Omega_{\max}$

由 distribution C 可看出: 分布越广越有可能出现 $\therefore \Omega_{\max} \rightarrow$ 宏观平衡 $\rightarrow$ 熵 $\max$
most spread out

假设 4 白 4 灰, 键能相同 (概率一致): $\Omega = \frac{N!}{n! \times (N-n)!}$



例: $\Omega_{4,0} = \frac{4!}{4! \times (4-4)!} = 1$ 灰同理

$$\Omega_{3,1} = \frac{4!}{3! \times (4-3)!} = 4$$

$$\Omega_{2,2} = \frac{4!}{2! \times (4-2)!} = 6$$

$$\Omega_{1,3} = \frac{4!}{1! \times (4-1)!} = 4$$

$$\Omega_{0,4} = 1$$

1. All white atoms on the left and all gray atoms on the right: $\Omega_{4,0} \times \Omega_{0,4} = 1$
2. Three white atoms on the left and one gray atom on the right: $\Omega_{3,1} \times \Omega_{1,3} = 16$
3. Two white atoms on the left and two gray atoms on the right: $\Omega_{2,2} \times \Omega_{2,2} = 36$
4. One white atom on the left and three gray atoms on the right is the same as three white atoms on the left and one gray atom on the right: $\Omega_{1,3} \times \Omega_{3,1} = \Omega_{3,1} \times \Omega_{1,3} = 16$
5. No white atoms on the left can only be done one way: $\Omega_{0,4} \times \Omega_{4,0} = 1$

$$\text{total } \Omega = \frac{8!}{4! \times 4!} = 70$$

no energetically favored bonds

(1) $A+B \rightarrow A+B$ at constant u.v. n

unmixed mixed

$$\Delta S_{conf} = S_{conf}(2) - S_{conf}(1)$$

$$= k_B \ln(\Omega_{conf}(2)) - k_B \ln(\Omega_{conf}(1)) = k_B \ln \frac{\Omega(2)}{\Omega(1)} = \text{最初1种构型}$$

$$\Omega_{conf}(2) = \frac{(n_a + n_b)!}{n_a! n_b!} \quad \text{个数, 即非 mole}$$

$$\Delta S'_{conf} = k_B \ln \frac{(n_a + n_b)!}{n_a! n_b!}$$

$$\Omega_{conf}(1) = 1$$

当 $n_k = 1 \text{ mol}$ 可用

Suppose $X_A = \frac{n_A}{n_A + n_B} \Rightarrow \Delta S_{conf} = -R(X_A \ln X_A + X_B \ln X_B)$ 用到 $\ln N! \approx N \ln N - N$

$\therefore$ 当 $X_A = X_B = 50\%$ (-边是AB各半) 时 ΔS_{max}

$$\Omega = \frac{n!}{n_0! n_1! n_2! \dots n_r!}$$

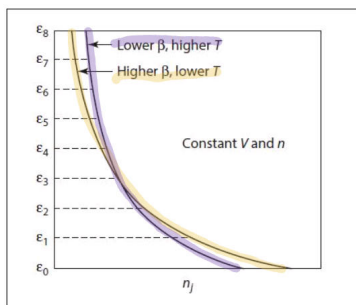
$\left\{ \begin{array}{l} 1 \sim r: n \text{ 种分类 (eg. } r \text{ 个能级, } r \text{ 个隔间...)} \\ n_r: \text{ 每个分类上的 atom 数} \end{array} \right.$

$$\Delta S = k_B \ln \Omega \quad \text{其中 } k_B = 1.381 \times 10^{-23} \text{ J/K}$$

$$\ln \Omega = n \ln n - n - \sum_{i=0}^{i=r} (n_i \ln n_i - n_i)$$

温度增加时, $\beta = \frac{1}{k_B T}$. $\beta \downarrow$:

高能级粒子 $\uparrow$



$$\ln \Omega_{total} \approx \ln \Omega_{max} = n \ln n - \sum_{i=0}^{i=r} n_i \ln n_i$$

$$\ln \Omega_{\text{total}} = n \ln P + \frac{U}{k_B T} \quad \text{其中 } P = \sum_{i=0}^r e^{-\beta \epsilon_i}$$

↓
常数

$$\delta \ln \Omega_{\text{total}} = \frac{\delta U}{k_B T} = \frac{\delta q_A}{k_B T} \quad (\text{等 } V)$$

$$\delta \ln \Omega_A \Omega_B = \delta \ln \Omega_A + \delta \ln \Omega_B = \frac{\delta q_A}{k_B T_A} + \frac{\delta q_B}{k_B T_B} = \frac{\delta q_A}{k_B} \left(\frac{1}{T_A} - \frac{1}{T_B} \right)$$

$$S_{\text{total}} = S_{\text{th}} \text{ 热熵} + S_{\text{conf}} \text{ 构型熵} + \dots$$

$$= k_B \ln \Omega_{\text{th}} + k_B \ln \Omega_{\text{conf}} = k_B \ln \Omega_{\text{th}} \Omega_{\text{conf}}$$

$$\Delta S_{\text{total}} = k_B \ln \frac{\Omega_{\text{th}(2)} \Omega_{\text{conf}(2)}}{\Omega_{\text{th}(1)} \Omega_{\text{conf}(1)}} \quad S_{\text{state 2}} - S_{\text{state 1}}$$

微观状态数

$$\left\{ \begin{array}{l} \Omega_{\text{th}}: \epsilon_1, \epsilon_2, \epsilon_3, \dots \\ \Omega_{\text{conf}}: \text{构型 (4 倍)} \\ \Omega_{\text{spin}}: \uparrow \downarrow \text{自旋 } 2^n \end{array} \right.$$

Chapter 5

- ① unmeasurable $\rightarrow$ measurable 建立新方程
- ② 考虑多种 composition 化学势 (chemical potential)
- ③ 考虑非体积功

$$\left. \begin{array}{l} U = q - w \\ q = U + w \\ \Delta H = (U_2 - U_1) + P(V_2 - V_1) \end{array} \right\} dH = \delta q_p = T \cdot ds$$

1. at constant T, V

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}} > 0$$

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} - \frac{\Delta U_{\text{system}}}{T} = +\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = -\frac{q_{\text{sys}}}{T}$$

$$-T \cdot \Delta S_{\text{total}} = \Delta U_{\text{sys}} - T \Delta S_{\text{sys}}$$

$$< 0 \quad \downarrow \text{自发条件: } \Delta U_{\text{system}} - T \Delta S_{\text{total}} \leq 0$$

$$\text{Defining: } A = U - TS$$

$$A_2 - A_1 = U_2 - U_1 - (T_2 S_2 - T_1 S_1) = q - w - (T_2 S_2 - T_1 S_1)$$

$$q_{\text{rev}} = T \Delta S = q + T \cdot \Delta S_{\text{irr}}$$

$$\therefore A_2 - A_1 \leq -w$$

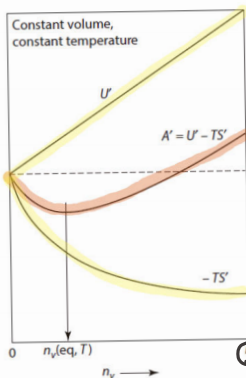
$$\delta W \leq T dS_{\text{sys}} - dU_{\text{sys}}$$

$$A_2 - A_1 + T \Delta S_{\text{irr}} = -w \quad (\text{等} T)$$

2. Constant T, P.

$$A_2 - A_1 + T \Delta S_{\text{irr}} = 0 \Rightarrow dA + T dS_{\text{irr}} = 0 \quad (\text{由于 } dS_{\text{irr}} > 0, \text{ 则 } dA < 0)$$

如果 reversible: 则 $dA = 0$ 平衡. $dA \leq 0$ 则自发



3. Gibbs free energy G 平衡时 G 最小

定义 $G = H - TS$

$$G_2 - G_1 = H_2 - H_1 - (T_2 S_2 - T_1 S_1) = (U_2 - U_1) + (P_2 V_2 - P_1 V_1) - (T_2 S_2 - T_1 S_1)$$

$$\therefore G_2 - G_1 = q - w + (P_2 V_2 - P_1 V_1) - (T_2 S_2 - T_1 S_1)$$

※ 等TP: $G_2 - G_1 = q - w + P(V_2 - V_1)$

$$w = w' + P(V_2 - V_1)$$

若只考虑 $w = Pdv$ 则 $\Delta G = q - w - T(S_2 - S_1) = q - T(S_2 - S_1) \leq 0$ 自发

※ 总结: } 等T, V, 无 $W_{\neq kv}$, $dA \leq 0$ 自发
 } 等T, P, 无 $W_{\neq kv}$, $dG \leq 0$ 自发

$$dU = TdS - PdV$$

定义

$$H = U + PV \Rightarrow dH = TdS + VdP$$

$$du = Tds - pdv, dA = du - Tds - sdT = -pdv - sdT$$

$$A = U - TS \Rightarrow dA = -SdT - PdV$$

$$G = H - TS \Rightarrow dG = -SdT + VdP$$

4. Chemical potential:

$$G = G(T, P, n_1, n_2, n_3, \dots) \text{ 多种 composition}$$

$$\therefore dG = \left(\frac{\partial G}{\partial T}\right)_{P, n_1, n_2, \dots} dT + \left(\frac{\partial G}{\partial P}\right)_{T, n_1, n_2, \dots} dP + \left(\frac{\partial G}{\partial n_i}\right)_{T, P, n_2, n_3, \dots} dn_i + \dots$$

$$dG = -SdT + VdP + \sum_{i=1}^k \left(\frac{\partial G}{\partial n_i}\right)_{T, P, n_{j \neq i}} dn_i$$

MA 意义: 无限大体系中, A 增加 1 mol, BCD ... 不变. TP 恒而有的 ΔG

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_1, \dots}$$

$$\therefore dG = -SdT + VdP + \sum_{i=1}^k \mu_i dn_i \quad \text{open systems}$$

$$dU = TdS - PdV + \sum_{i=1}^{i=k} \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_j, i \neq j} dn_i$$

$$dH = TdS + VdP + \sum_{i=1}^{i=k} \left(\frac{\partial H}{\partial n_i} \right)_{S, P, n_j, i \neq j} dn_i$$

$$dA = -SdT - PdV + \sum_{i=1}^{i=k} \left(\frac{\partial A}{\partial n_i} \right)_{T, V, n_j, i \neq j} dn_i$$

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_j, i \neq j} = \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_j, i \neq j} = \left(\frac{\partial H}{\partial n_i} \right)_{S, P, n_j, i \neq j} = \left(\frac{\partial A}{\partial n_i} \right)_{T, V, n_j, i \neq j}$$

5. Thermodynamic relations

利用 4 个式子:

$$\begin{cases} dU = Tds - pdv & \rightarrow T = \left(\frac{\partial U}{\partial s} \right)_v & P = - \left(\frac{\partial U}{\partial v} \right)_s \\ dH = Tds + vdp & \rightarrow T = \left(\frac{\partial H}{\partial s} \right)_p & V = \left(\frac{\partial H}{\partial p} \right)_s \\ dA = -SdT - PdV & \rightarrow S = - \left(\frac{\partial A}{\partial T} \right)_V & P = - \left(\frac{\partial A}{\partial V} \right)_T \\ dG = -SdT + Vdp & \rightarrow S = - \left(\frac{\partial G}{\partial T} \right)_P & V = \left(\frac{\partial G}{\partial P} \right)_T \end{cases}$$

6. Maxwell's equations:

有关数学: $z = z(x, y) \rightarrow dz = \underbrace{\left(\frac{\partial z}{\partial x} \right)_y}_{L(x, y)} dx + \underbrace{\left(\frac{\partial z}{\partial y} \right)_x}_{M(x, y)} dy$

$$\therefore dz = Ldx + Mdy$$

再求导: $\left(\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)_y \right)_x = \left(\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)_y \right)_x \quad \left(\frac{\partial L}{\partial x} \right)_y = \left(\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)_x \right)_y$

则: $\left(\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)_y \right)_x = \left(\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)_x \right)_y = \frac{\partial^2 z}{\partial x \partial y} \quad \text{则} \left(\frac{\partial L}{\partial y} \right)_x = \left(\frac{\partial M}{\partial x} \right)_y$

$$dU = Tdx - PdV \Rightarrow \left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial P}{\partial S} \right)_V$$

$$dH = TdS + VdP \Rightarrow \left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_P$$

$$dA = -SdT - PdV \Rightarrow \left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial P}{\partial T} \right)_V$$

$$dG = -SdT + VdP \Rightarrow \left(\frac{\partial S}{\partial P} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_P$$

Maxwell relations can be obtained by considering

$$\left\{ \begin{aligned} \left(\frac{\partial V}{\partial T} \right)_P &= \alpha_V \quad (\alpha: \text{热膨胀系数}) \\ \left(\frac{\partial V}{\partial P} \right)_T &= -V\beta_T \quad (\beta_T: \text{压缩系数}) \end{aligned} \right.$$

7. the first TdS Equation:

$$S = S(T, V)$$

$$ds = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV$$

$$Tds = dq_{rev} = du = C_V dT \rightarrow \left(\frac{\partial S}{\partial T} \right)_V = \frac{C_V}{T}$$

$$\therefore ds = \frac{C_V}{T} dT + \left(\frac{\partial P}{\partial T} \right)_V dV$$

$$= \frac{\alpha}{\beta_T} dV$$

$$\text{综上: } Tds = C_V dT + T \cdot \frac{\alpha}{\beta_T} dV$$

• For an isothermal expansion

$$\Delta S = \int \frac{\alpha}{\beta_T} dV > 0 \quad C_V dT = 0$$

• If the temperature is raised at constant volume

$$\Delta S = \int \frac{C_V}{T} dT > 0 \quad \frac{\alpha}{\beta_T} = 0$$

• For an isentropic expansion

$$0 = C_V dT + T \frac{\alpha}{\beta_T} dV \Rightarrow \left(\frac{\partial T}{\partial V} \right)_S = - \frac{T\alpha}{C_V \beta_T} < 0$$

若为 1mol ideal gas:

$$\left(\frac{\partial P}{\partial T} \right)_V = \frac{R}{V}$$

$$\therefore ds = \frac{C_V}{T} dT + \frac{R}{V} dV$$

$$S_2 - S_1 = C_V \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}$$

The second Tds Equation : $S = S(T, P) \rightarrow Tds = C_p dT - \alpha V dp$

$$ds = \left(\frac{\partial s}{\partial T}\right)_P dT + \left(\frac{\partial s}{\partial P}\right)_T dP$$

$$\left. \begin{aligned} ds &= \frac{C_p}{T} dT - \alpha V dP \\ dV &= \alpha V dT - V \beta_T dP \end{aligned} \right\}$$

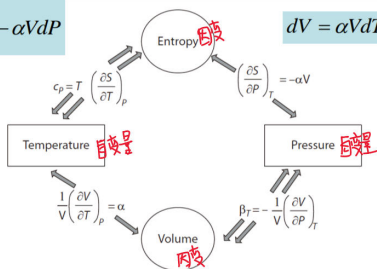
$$Tds = q, ds = \frac{q}{T} = C_p \cdot \left(\frac{\partial s}{\partial T}\right)_P = \frac{C_p}{T}$$

$$dS = \frac{C_p}{T} dT - \alpha V dP$$

$$dV = \alpha V dT - V \beta_T dP$$

$$\left(\frac{\partial s}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

$$= -\alpha V$$



第五章总结:

1. 定义式:

$$U = q - w$$

$$H = U + pV$$

$$A = U - TS$$

$$G = H - TS$$

2. 热力学等式: $du = Tds - pdv$

$$dh = Tds + vdp$$

$$da = -sdt - pdv$$

$$dg = -sdt + vdp$$

3. thermodynamic relation

4. maxwell 等式.

$$5. \left. \begin{aligned} S = S(T, V) &, ds = \frac{C_v}{T} dT + \frac{P}{V} dV \\ S = S(T, P) &, ds = \frac{C_p}{T} dT - \alpha V dP \end{aligned} \right\}$$

$$\textcircled{2} \left\{ \begin{array}{l} S = S(T, P) \quad , \quad ds = \frac{C_p}{T} dT - \alpha v dp \\ V = V(T, P) \quad dv = \alpha v dT - \beta v dp \end{array} \right.$$

$$b. \left\{ \begin{array}{l} \frac{d(\Delta A/T)}{dT} = -\frac{\Delta U}{T^2} \\ \frac{d(\Delta G/T)}{dT} = -\frac{\Delta H}{T^2} \end{array} \right.$$

chapter 6

$C_p, C_v \rightarrow$ 和 T 有关 $f(T)$

$$C_v = \left(\frac{\partial U}{\partial T} \right)_v \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

$$du = C_v dT \quad dH = C_p dT \quad \Delta H = H(T, P) - H(T_0, P) = \int_{T_0}^{T_2} C_p dT$$

$\textcircled{1} \epsilon_i = (i + \frac{1}{2}) h\nu$ 能级 first calculation by Einstein

$\downarrow U = 3 \sum n_i \epsilon_i$ 斜振子有3个方向

partition function: $p = \bar{z}_i \exp\left(-\frac{\epsilon_i}{k_B T}\right)$, $n_i = \frac{n \cdot \exp\left(-\frac{\epsilon_i}{k_B T}\right)}{p}$

$$U = 3 \sum n h\nu \left(i + \frac{1}{2} \right) \exp\left(-\frac{\epsilon_i}{k_B T}\right) / \sum_i \exp\left(-\frac{\epsilon_i}{k_B T}\right)$$

Thermodynamics of Materials & Phase transformations

$$U = \frac{3}{2} n h\nu + \frac{3 n h\nu}{\exp\left(\frac{h\nu}{k_B T}\right) - 1}$$

$\downarrow C_p f(T) = \left(\frac{\partial U}{\partial T} \right)_p$

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \frac{\exp\left(-\frac{\theta_E}{T}\right)}{\left[\exp\left(-\frac{\theta_E}{T}\right) - 1 \right]^2} \quad \text{其中 } \theta_E = \frac{h\nu}{k_B}$$

然而, Debye 认为: ν 有范围. $\nu_{\max} = \frac{\nu}{\lambda_{\min}} = 10^{13} \text{ s}^{-1}$

有在微分积分

$$c_V = \frac{9nh^3}{k_B^2 \theta_D^3} \int_0^{\theta_D} \nu^2 \left(\frac{h\nu}{k_B T} \right)^2 \frac{\exp\left(-\frac{h\nu}{k_B T}\right)}{\left[\exp\left(-\frac{h\nu}{k_B T}\right) - 1\right]^2} d\nu \quad \text{其中 } x = \frac{h\nu}{k_B T} \quad c_V = 9R \left(\frac{T}{\theta_D} \right) \int_0^{\theta_D/T} x^4 \frac{e^{-x}}{[e^{-x} - 1]^2} dx$$

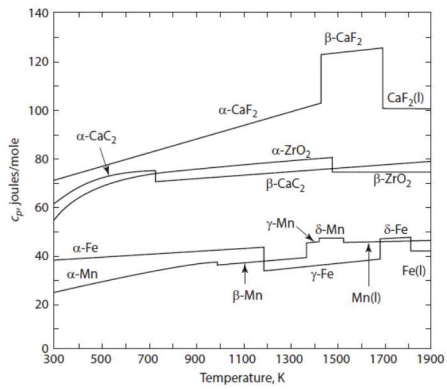
$$V_D = V_{max} - \theta_D = \frac{h\nu_D}{k_B}$$

$$c_V = 9R \times 25.98 \left(\frac{T}{\theta_D} \right)^3 = 1944 \left(\frac{T}{\theta_D} \right)^3 \quad (\text{integral} = 25.98)$$

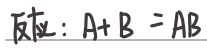
$$\therefore C_V = 8T + 1943 \left(\frac{T}{\theta_D} \right)^3$$

2. C_p 一般表现形式:

$$C_p = a + bT + cT^2 \quad \text{abc查表. 且 } T \text{ 相变点时 } C_p \text{ 突变}$$

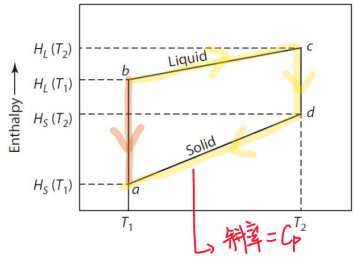


$$\Delta H = H(T_2, P) - H(T_1, P) = \int_{T_1}^{T_2} C_p dT$$



$$\Delta H = H_{AB} - [H_A + H_B]$$

例: 已知平衡条件 $\Delta H (c \rightarrow d)$, 求任一条件下 $\Delta H (b \rightarrow a)$



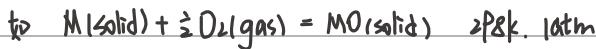
$$\Delta H (b \rightarrow a) = \Delta H (b \rightarrow c) + \Delta H (c \rightarrow d) + \Delta H (d \rightarrow a)$$

$$\Delta H (l \rightarrow s, T_1) = H(s, T_1) - H(l, T_1)$$

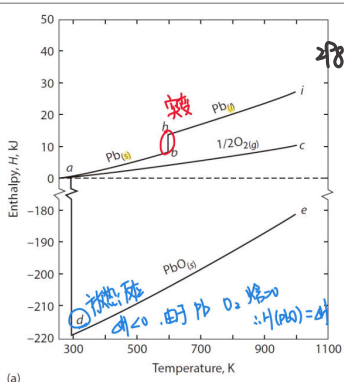
$$\rightarrow \Delta H (l \rightarrow s, T_1) = \int_{T_1}^{T_2} c_p(l) dT + \Delta H (l \rightarrow s, T_2) + \int_{T_2}^{T_1} c_p(s) dT$$

if $C_p(l) - C_p(s) = 0$ 则 $\Delta H_{T_1} = \Delta H_{T_2}$

反义 $H=0$ 的点: 标准状态 (298K, 1atm) 下的单质



$$\Delta H_{298} = H_{MO(s)298} - \underbrace{H_{M(s)298}}_{=0} - \underbrace{\frac{1}{2} H_{O_2298}}_{=0}$$



(a)

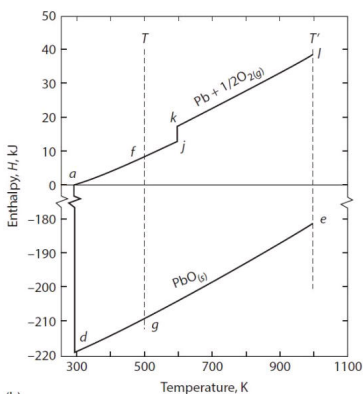
298 ≤ T ≤ 600 时: $H_{Pb(s)} = \int_{298}^T C_{p,Pb(s)} dT$

O₂: 298 ≤ T ≤ 3000 K 时:

$$\frac{1}{2} H_{O_2(g)} T = \frac{1}{2} \int_{298}^T C_{p,O_2(g)} dT$$

PbO(s): 298K ≤ T ≤ 1159

$$H_{PbO(s),T} = \underbrace{-21000}_{298K \text{ 时}} + \int_{298}^T C_{p,PbO(s)} dT$$



(b)

$$\begin{aligned} \Delta H_{PbO,298K} &= \Delta H_{(a \rightarrow f)} + \Delta H_{(f \rightarrow g)} + \Delta H_{(g \rightarrow d)} \\ &= \int_{298}^T \left(\frac{1}{2} C_{p,O_2(g)} + C_{p,Pb(s)} \right) dT + \Delta H_{PbO,T} + \int_{298}^T C_{p,PbO(s)} dT \end{aligned}$$

反应物 生成物

$$\Delta H_{298K} = \Delta H(a \rightarrow d) = \Delta H(a \rightarrow j) + \Delta H(j \rightarrow k) + \Delta H(k \rightarrow l)$$

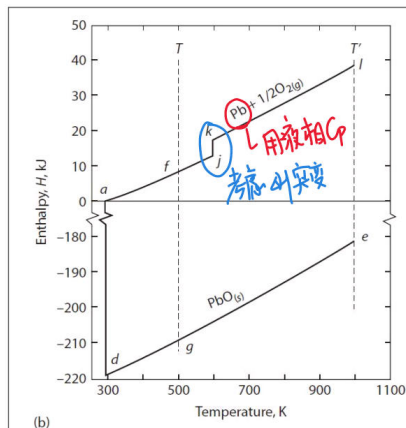
$$\Delta H(l \rightarrow e) + \Delta H(e \rightarrow g) + \Delta H(g \rightarrow d)$$

$$\Delta H(a \rightarrow j) = \int_{298K}^{T_{m,Pb}} \left(C_{p,Pb(s)} + \frac{1}{2} C_{p,O_2(g)} \right) dT$$

$$\Delta H(j \rightarrow k) = \text{the latent heat of melting of Pb at } T_{m,Pb} = 4810 \text{ J}$$

$$\Delta H(h \rightarrow l) = \int_{T_{m,Pb}}^{298K} \left(C_{p,Pb(l)} + \frac{1}{2} C_{p,O_2(g)} \right) dT$$

$$\Delta H(l \rightarrow e) = \Delta H_T$$



(b)

$$3. \Delta H(T_2) - \Delta H(T_1) = \int_{T_1}^{T_2} \Delta C_p dT \quad \text{Kirchhoff's law}$$

$$\Delta H_T = H_{298} + \int_{298}^T C_p dT$$

$$S(T_2, P) - S(T_1, P) = \int_{T_1}^{T_2} \frac{C_p dT}{T} \quad (1 \text{ mol})$$

$$S_T = S_0 + \int_0^T \frac{C_p dT}{T} \quad S=0 \text{ 条件: } T=0\text{K, 稳定均匀 (可为非单质)}$$

热力学第三定律内容: 熵在温度 $\rightarrow$ 绝对零度时趋于定值。对于完整晶体定值为零

$$\Delta G = \Delta H - T \cdot \Delta S$$

$$\text{for reaction: } \text{Pb} + \frac{1}{2} \text{O}_2 = \text{PbO}, \quad \Delta S_T = S_{T, \text{PbO}} - S_{T, \text{Pb}} - \frac{1}{2} S_{T, \text{O}_2}$$

4. 研究 P 如何影响 $\Delta G, \Delta S$:

$$\textcircled{1} dH = T ds + V dp \rightarrow \frac{dH}{dp} = T \frac{ds}{dp} + V \rightarrow \left(\frac{\partial H}{\partial p} \right)_T = T \left(\frac{\partial S}{\partial p} \right)_T + V$$

$-\left(\frac{\partial V}{\partial T} \right)_p$ 根据 maxwell

$$\therefore \left(\frac{\partial H}{\partial p} \right)_T = -T \left(\frac{\partial V}{\partial T} \right)_p + V = -T \alpha V + V = V(1 - \alpha T)$$

$$\therefore H(P_2, T) - H(P_1, T) = \int_{P_1}^{P_2} V(1 - \alpha T) dp \quad \text{对 ideal gas, } p \text{ 不引发 } \Delta H$$

$$\textcircled{2} ds = \left(\frac{\partial S}{\partial p} \right)_T dp$$

$$\left(\frac{\partial S}{\partial p} \right)_T = - \left(\frac{\partial V}{\partial T} \right)_p \rightarrow ds = - \left(\frac{\partial V}{\partial T} \right)_p dp = -\alpha V dp$$

$$\therefore \Delta S = S(P_2, T) - S(P_1, T) = - \int_{P_1}^{P_2} \alpha V dp \quad \text{ideal gas: } \alpha = \frac{1}{T}$$

$$\Delta H = H(T_2, P) - H(T_1, P) = \int_{T_1}^{T_2} c_p dT$$

$$\Delta H = H(P_2, T) - H(P_1, T) = \int_{P_1}^{P_2} V(1 - \alpha T) dP$$

$$\Delta S = S(P_2, T) - S(P_1, T) = - \int_{P_1}^{P_2} \alpha V dP$$

$$\Delta S = S(T_2, P) - S(T_1, P) = \int_{T_1}^{T_2} \frac{c_p dT}{T}$$

★ 高温下 $C_v \approx n \cdot 3R$ (n 为原子数. eg. Fe_2SO_4 中 $n=7$)

chapter 8:

1. Dalton's Law of Partial Pressures

1) 体积 V' 含有 n_A 的 A: $P_A = \frac{n_A RT}{V'}$ + n_B 的 B $\rightarrow P_A + P_B = (n_A + n_B) \frac{RT}{V'}$

$$\therefore \frac{P_A}{P} = x_A = \frac{n_A}{n_{\text{总}}} \quad P_A = x_A P$$

12) $\bar{Q}_i = \left(\frac{\partial Q'}{\partial n_i} \right)_{T, P, n_j, n_k, \dots}$ → 任一热力学变量
↓
 rate of Q' = partial molar Q'

$$\bar{G}_i = \left(\frac{\partial G'}{\partial n_i} \right)_{T, P, n_j, \dots} = M_i \quad \bar{G}_i: \text{partial molar Gibbs free energy}$$

2. 理想气体的混合焓:

for each component gas in mixture gases 中: $\bar{G}_i = G_i^\circ + RT \ln P_i = G_i^\circ + RT \ln x_i + RT \ln P$

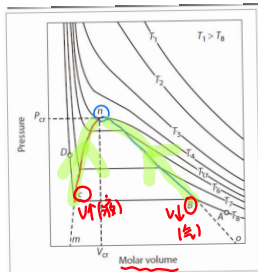
$$\left(\frac{\partial \bar{G}_i}{\partial T} \right)_{P, n_i} = \left(\frac{\partial G_i^\circ}{\partial T} \right)_{P, n_i} \quad \swarrow \text{同除 } T \text{ 再对 } T \text{ 偏导}$$

$$\underline{-\frac{H_i}{T^2}} = \underline{-\frac{H_i^\circ}{T^2}} \quad G = H - TS$$

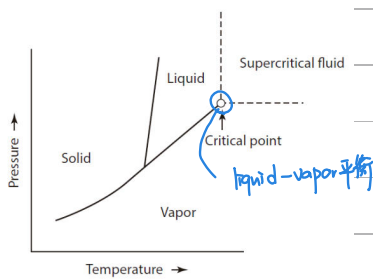
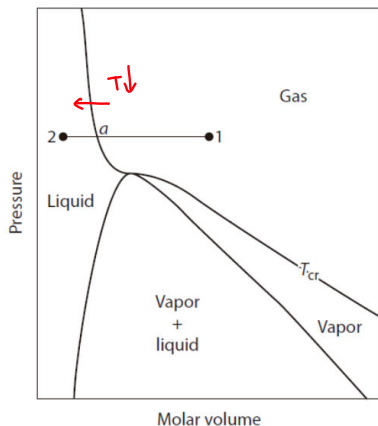
其中 H_i : 混合后气体的焓; H_i° : 混合前气体的焓

state c: 完全冷凝 (condensation)

$C \rightarrow D$ 等温 $p \uparrow$



T_{cr} 时 $V_{liquid} = V_{gas}$



5. The Van Der Waals Fluid:

* real gas 存在体积 * real gas 之间有相互作用力

$$\left(p + \frac{a}{v^2}\right)(v-b) = RT \quad Z = \frac{pV_{cr}}{nRT_{cr}}$$

$$\begin{aligned} P_{cr} &= \frac{RT_{cr}}{(V_{cr}-b)} - \frac{a}{V_{cr}^2} \\ \left(\frac{\partial P}{\partial V}\right)_{T_{cr}} &= -\frac{RT_{cr}}{(V_{cr}-b)^2} + \frac{2a}{V_{cr}^3} = 0 \\ \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_{cr}} &= \frac{2RT_{cr}}{(V_{cr}-b)^3} - \frac{6a}{V_{cr}^4} = 0 \end{aligned}$$



$$\begin{aligned} T_{cr} &= \frac{8a}{27bR} \\ V_{cr} &= 3b \\ P_{cr} &= \frac{a}{27b^2} \end{aligned}$$

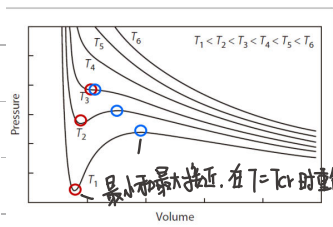
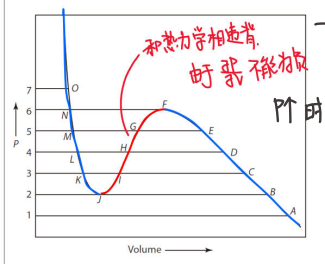


Figure 8.7 The isothermal variation of V with P for a van der Waals fluid at several temperatures.

a b 值可查表



→ $T < T_{cr}$ 时.

PT 时 V 应 ↓ ∴ Curve is a region of instability

$$dG = -SdT + Vdp$$

↓ T 不变

$$dG = Vdp \quad \therefore \Delta G = \int_{P_A}^P Vdp, \quad G = G_A + \int_{P_A}^P Vdp$$

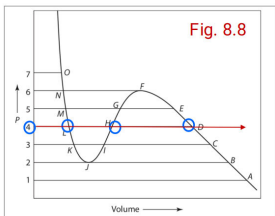


Fig. 8.8

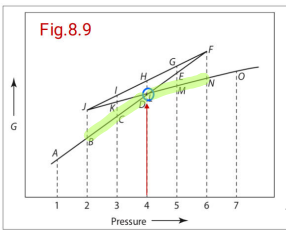


Fig.8.9

→ $P_2 - P_6$ 有 3 个相.

Figure 8.8 the isothermal $P \sim V$ curve of a van der Waals fluid at a $T < T_{cr}$.

Figure 8.9 the isothermal $G \sim P$ curve of a van der Waals fluid at a $T < T_{cr}$.

eg. P_3 点是 1, k, c 个相

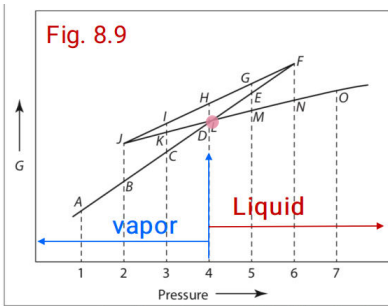


Fig. 8.9

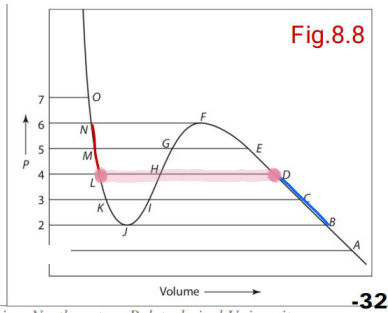


Fig.8.8

at P_4 : G_D (vapor phase) = G_L (liquid phase)

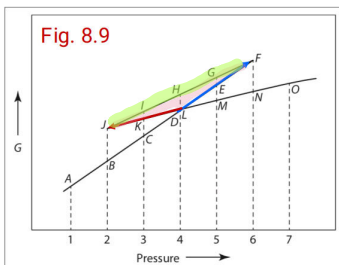


Fig. 8.9

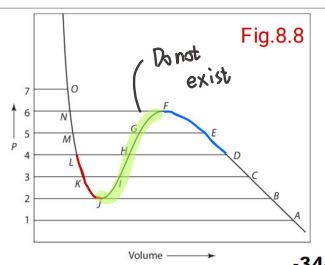
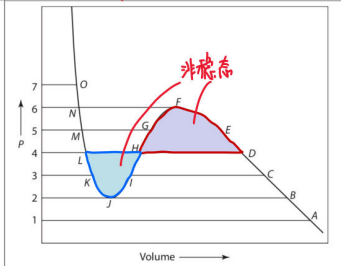
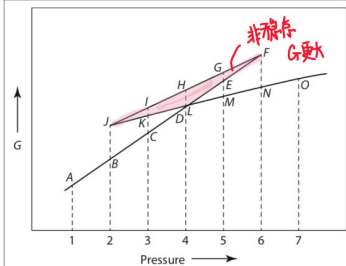


Fig.8.8



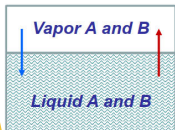
Chapter 9

二元溶液

1. Raoult's law: (针对溶剂)

$$P_A = x_A P_A^*$$

↓ saturated vapor pressure of pure liquid A
Vapor pressure



A binary solution A-B

Rate of evaporation: $Y_e(A)$

Rate of condensation: $Y_c(A)$

$$Y_e(A) = k_A^* P_A^* = Y_c(A)$$

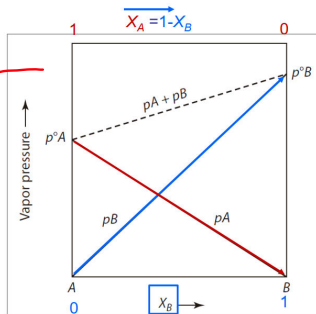


Figure 9.1 the vapor pressures of a binary solution. Both components of this solution obey Raoult's law at all compositions.

将少量的B液体加入A液体

$$Y_e(A) x_A = k_A^* P_A^*$$

$$P_A = x_A \cdot P_A^*$$

$$Y_e(A) = k_A^* P_A^*$$

前提: $E_{A-A} = E_{A-B} = E_{B-A}$ energy band 相同

2. Henry's law: (针对溶质)

在溶剂中加入一定量的溶质B: B atom 需克服很大的energy barriers to evaporate.

∴ the evaporation rate ↓: $Y_e(B) \rightarrow Y_e'(B)$

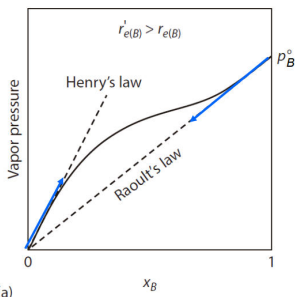
$$r'_{e(B)} < r_{e(B)}$$



$$k_B < 1$$

$$Y_e(B) = k_B^* P_B^* \rightarrow P_B = \frac{Y_e(B)}{k_B^*} x_B P_B^* \rightarrow P_B = k_B x_B$$

$$Y_e'(B) x_B = k_B^* P_B^*$$



$$p_B = X_B p_B^{\circ} \quad \text{Raoult's law,}$$

↓
B 最有利
B 最多

压力加 常数 + 理想时 B 最多
= 1 如是溶液溶质

$$p_B = k_B^{\circ} X_B \quad \text{Henry's law,}$$

3. the Thermodynamic Activity:

ideal gas: $G = G^{\circ} + RT \ln p$

nonideal gas: $dG = RT d \ln f \quad (f: \text{Fugacity})$

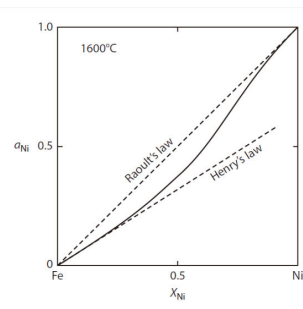
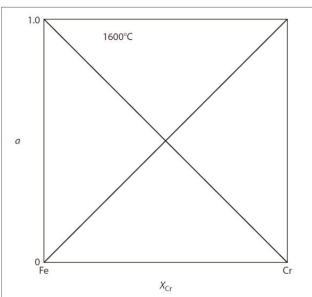
$G = G^{\circ} + RT \ln f \quad p > 0, f/p \rightarrow 1 \quad \text{无限小压力无限大体积}$

activity of $i \equiv a_i = \frac{f_i}{f_i^{\circ}} \rightarrow T \text{ 时 component } i \text{ 的 fugacity}$
 $\rightarrow T \text{ 时 pure } i \text{ 的 fugacity (标准状态下)}$

对于理想 solution 来说: $f_i = p_i \therefore a_i = \frac{p_i}{p_i^{\circ}} \xrightarrow{\text{Raoult's law}} \frac{p_i}{p_A = x_A p_A^{\circ}} \rightarrow a_i = x_i$
 混合物则 $a_i = 1$

T 高到一定时, 活度线性取决于 x_A

$a_i = \frac{p_i}{p_i^{\circ}} \xrightarrow{\text{Henry's law}} \frac{p_i}{p_i = k_i^{\circ} x_i} \rightarrow a_i = \frac{k_i x_i}{p_i^{\circ}} = k_{(i)} x_i$



Ni 和 Fe 更易分开. $E_{\text{Ni-Fe}} < E_{\text{Ni-Ni}}$ 键能更负

4. The Gibbs - Duhem Equation:

$$\bar{Q}' = \bar{Q}'(T, p, n_i, n_j, \dots)$$

the variation of $\bar{Q}'$ with the composition of the solution:

$$d\bar{Q}' = \left[\frac{\partial \bar{Q}'}{\partial n_i} \right]_{T, p, n_j, \dots} + \dots = \bar{Q}_i dn_i + \bar{Q}_j dn_j + \dots$$

the value of $\bar{Q}$ per mole of i in the solution

$$\therefore \bar{Q}' = n_i \bar{Q}_i + n_j \bar{Q}_j + \dots$$

↓ 求导

$$d\bar{Q}' = n_i d\bar{Q}_i + n_j d\bar{Q}_j + n_k d\bar{Q}_k + \dots + \bar{Q}_i dn_i + \bar{Q}_j dn_j + \bar{Q}_k dn_k + \dots$$

$$d\bar{Q}' = \bar{Q}_i dn_i + \bar{Q}_j dn_j + \bar{Q}_k dn_k + \dots$$

$$\underbrace{n_i d\bar{Q}_i + n_j d\bar{Q}_j + n_k d\bar{Q}_k + \dots}_{\text{同除 } n} = 0 \quad \therefore \sum n_i d\bar{Q}_i = 0$$

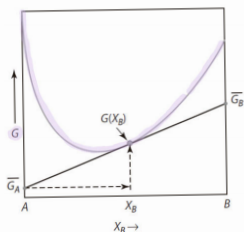
$$\therefore \sum x_i d\bar{Q}_i = 0$$

$$X_A d\bar{G}_A + X_B d\bar{G}_B = 0$$

$$\underbrace{\bar{G}'}_{\text{total } G \text{ of the solution}} = n_A \bar{G}_A + n_B \bar{G}_B \xrightarrow{\text{同除 } n_A + n_B}$$

$$\bar{G} = x_A \bar{G}_A + x_B \bar{G}_B$$

molar $\bar{G}$



$$\bar{G}_A = G - X_B \frac{dG}{dX_B}$$

$$\bar{G}_B = G + X_A \frac{dG}{dX_B}$$

3个等温步骤:

T, P 恒定
(a) 1 mole pure i evaporate at p_i : constant T, p , $\Delta G(a) = 0$

(b) 降低压强: $p_i^0 \rightarrow p_i$: $\Delta G(b) = G(p_i, T) - G(p_i^0, T) = RT \ln \frac{p_i}{p_i^0}$

(c) 1 mole vapor i 在 p_i 下冷凝: $\Delta G(c) = 0$

$\therefore$ difference between molar $\bar{G}$ in the solution and pure $i = \Delta G(a) + \Delta G(b) + \Delta G(c)$

$$\Delta G_i(p) = RT \ln \frac{P_i}{P_i^\circ}, \quad a = \frac{P_i}{P_i^\circ} \Rightarrow \Delta G_i(p) = G_i(\text{in solution}) - G_i^\circ(\text{pure}) = RT \ln a_i$$

$$\Delta \bar{G}_i^M = \bar{G}_i - G_i^\circ = RT \ln a_i$$

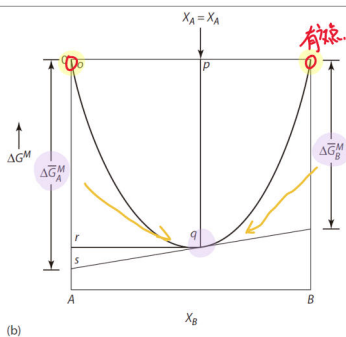
Solution 中 加入 1 mole i 时的 G 变化

The Gibbs free energy before mixing $= n_A G_A^\circ + n_B G_B^\circ$ ①

$$\Delta G^M = ② - ①$$

The Gibbs free energy after mixing $= n_A \bar{G}_A + n_B \bar{G}_B$ ② $= RT(n_A \ln a_A + n_B \ln a_B)$

for 1 mole: $\Delta G^M = RT(X_A \ln a_A + X_B \ln a_B)$



→ 该温度下 B 完全溶于 A; A 完全溶于 B

1mol: $\Delta G^M = RT(X_A \ln a_A + X_B \ln a_B)$

↓ ideal ($a_i = x_i$)

$$\Delta G^{M, id} = RT(X_A \ln X_A + X_B \ln X_B)$$

$$\Delta \bar{G}_A^{M, id} = RT \ln X_A = RT \ln a_A$$

$$\Delta \bar{G}_B^{M, id} = RT \ln X_B = RT \ln a_B$$

$$\left[\frac{\partial (\bar{G}_i - G_i^\circ)}{\partial P} \right]_{T, \text{comp}} = (\bar{V}_i - V_i^\circ) \Rightarrow \left[\frac{\partial \bar{G}_i^M}{\partial P} \right]_{T, \text{comp}} = \Delta \bar{V}_i^M, \quad \Delta \bar{V}^M = n_A \Delta \bar{V}_A^M + n_B \Delta \bar{V}_B^M$$

↓ ideal

$$\Delta \bar{G}_i^{M, id} = RT \ln x_i \text{ (和 } P \text{ 无关)} \Rightarrow \Delta \bar{V}_i^{M, id} = 0$$

∴ ideal solution $V = \text{sum of pure components } V$.

$$\textcircled{1} \left[\frac{\partial(\frac{\Delta \bar{G}_i^M}{T})}{\partial T} \right]_{P, \text{comp}} = - \frac{\Delta \bar{H}_i^M}{T^2}$$

$$\text{∴ } \Delta \bar{G}_i^M, \text{id} = RT \ln x_i \rightarrow \boxed{\frac{d(R \ln x_i)}{dT}} = - \frac{\Delta \bar{H}_i^M, \text{id}}{T^2} \Rightarrow \Delta \bar{H}_i^M, \text{id} = 0 = \bar{H}_i - H_i^0$$

和 T 无关 ∴ = 0

$$\Delta H^M = n_A \Delta \bar{H}_A^M + n_B \Delta \bar{H}_B^M \xrightarrow{\text{同除 } (n_A + n_B)} \Delta H^M = x_A \Delta \bar{H}_A^M + x_B \Delta \bar{H}_B^M = 0$$

∴ for ideal solution: $\Delta H^M, \text{id} = 0$

$$\textcircled{2} \left(\frac{\partial \Delta \bar{G}^M}{\partial T} \right)_{P, \text{comp}} = -\Delta \bar{S}^M \xrightarrow{\Delta \bar{G}^M, \text{id} = RT(x_A \ln x_A + x_B \ln x_B)} \Delta \bar{S}^M, \text{id} = -R(x_A \ln x_A + x_B \ln x_B)$$

和 T 无关

$$\text{total } \Delta \bar{S}^M = -k_B N_0 (n_A \ln x_A + n_B \ln x_B)$$

$$\rightarrow \Delta \bar{S}^M = -R(n_A \ln x_A + n_B \ln x_B)$$

1.38 × 10⁻²³ 6.02 × 10²³

$$\Delta \bar{S}^M (\text{1 mol}) = -R(x_A \ln x_A + x_B \ln x_B)$$

ideal) $\Delta \bar{G}_A^M, \text{id} = RT \ln x_A, \Delta \bar{S}_A^M, \text{id} = -R \ln x_A$

$\Delta \bar{G}_B^M, \text{id} = RT \ln x_B, \Delta \bar{S}_B^M, \text{id} = -R \ln x_B$

∴ for ideal solution: $\Delta \bar{G}^M, \text{id} = -T \Delta \bar{S}^M, \text{id} < 0 \quad (\Delta \bar{H}^M, \text{id} = 0)$

9.7. Nonideal Solutions:

Activity coefficient, $\gamma_i = \frac{a_i}{x_i}$ } $\gamma_i = 1$ Raoultian 行为

和 T 有关

升温 $\gamma > 1$

ideal: $\gamma_i = 1$ ∴ 无限 solution 趋于理想化

$\gamma_i > 1$ 和 Raoultian behavior 正偏差

负

$$a_i = \gamma_i x_i$$

$$\bar{\Delta G}_i^M = RT \ln a_i = RT \ln \gamma_i + RT \ln X_i$$

↓

$$\frac{\partial(\bar{G}_i^M/T)}{\partial T} = \frac{\partial(R \ln \gamma_i)}{\partial T} = -\frac{\Delta \bar{H}_i^M}{T^2} \xrightarrow{d(\frac{1}{T}) = \frac{dT}{-T^2}} \frac{\partial(R \ln \gamma_i)}{\partial(\frac{1}{T})} = \Delta \bar{H}_i^M$$

$$\Delta \bar{H}^M (\text{per mole}) = X_A \Delta \bar{H}_A^M + X_B \Delta \bar{H}_B^M$$

$\Delta \bar{H}^M > 0$ endothermic 吸热 $\Delta \bar{H}^M < 0$ exothermic

$X_i > 1$ eg: Fe-Cu 正偏析 positive deviation

↳ 一般吸热, $E_{A-B} > E_{A-A}, E_{B-B}$ ∴ 更易形成 A-A B-B (双相); 而 A-B (互溶体)

$X_i < 1$ eg: Fe-Ni 负偏析 $\Delta \bar{H}_i^M < 0$ $E_{A-B} < E_{A-A}, E_{B-B}$

更易形成 ordered phase 或 compound

9.8 Gibbs-Duhem Relation 的应用:

Gibbs-Duhem equation: $\sum X_i d\bar{Q}_i = 0$

$$\therefore X_A d\bar{G}_A^M + X_B d\bar{G}_B^M = 0 \xrightarrow{\bar{\Delta G}_i^M = RT \ln a_i} X_A d \ln a_A + X_B d \ln a_B = 0$$

$$\rightarrow da_A = -\frac{X_B}{X_A} d \ln a_B, \quad d \log a_A = -\frac{X_B}{X_A} d \log a_B$$

$$\therefore \log a_A \text{ at } X_A = - \int_{\log a_B \text{ at } X_A=1}^{\log a_B \text{ at } X_A=X_A} \left(\frac{X_B}{X_A} \right) d \log a_B$$

* Henry's law for solute B (溶质): $a_i = k_{(i)} X_i$

$$a_B = \gamma_B X_B \rightarrow \ln a_B = \ln \gamma_B + \ln X_B \rightarrow d \ln a_B = d \ln \gamma_B \quad \xrightarrow{d \ln a_B = -\frac{X_B}{X_A} d \ln a_A} d \ln a_A = d \ln X_A$$

$$\Rightarrow \ln a_A = \ln X_A + \ln(\text{constant})$$

$$a_A = \text{constant} \cdot X_A$$

* 两 component (A, B) 溶液:

$$\Delta \bar{G}^M = X_B \int_0^{X_A} \frac{\Delta \bar{G}_A^M}{X_B^2} dX_A \xrightarrow{\Delta \bar{G}_i^M = RT \ln a_i} \Delta \bar{G}^M = RT X_B \int_0^{X_A} \frac{\ln a_A}{X_B^2} dX_A$$

In the Fe-Ni system (Figure 9.8)

$$\Delta G^M = RT X_{Fe} \int_0^{X_{Ni}} \frac{\ln a_{Ni}}{X_{Fe}^2} dX_{Ni}$$

✓ In the Cu-Fe system (Figure 9.9)

$$\Delta G^M = RT X_{Fe} \int_0^{X_{Cu}} \frac{\ln a_{Cu}}{X_{Fe}^2} dX_{Cu}$$

$$\alpha_i = \frac{\ln f_i}{(1-x_i)^2}$$

7.9 Regular solutions (以下公式前提)

ideal solution: $\alpha_i = x_i$ $\Delta \bar{H}_i^M = 0$ $\Delta \bar{V}_i^M = 0$ $\Delta \bar{S}_i^M = R \ln x_i$

什么叫 regular solution: 满足 $\Delta \bar{H}^M = \alpha x_A x_B$ 且 $\Delta \bar{S}^M = \Delta \bar{S}^{M, id}$

$$\rightarrow \Delta \bar{G}^M = \alpha x_A x_B - T \Delta \bar{S}^{M, id}$$

Excess molar Gibbs free energy: $G_A^{XS} = \Delta \bar{G}^M - \Delta \bar{G}^{M, id} = \Delta \bar{H}^M = \alpha x_A x_B$

for regular solution: $\frac{\partial G_A^{XS}}{\partial T} = \frac{\partial (\Delta \bar{H}^M - T \Delta \bar{S}^M)}{\partial T} = -S_A^{XS} = 0$ $G_A^{XS} = \alpha x_B^2$
 $G_B^{XS} = \alpha \cdot x_A^2$

for any composition: $\bar{G}_A^{XS} = RT \ln \gamma_A$ Ideal 和 real 的偏移

$$\bar{G}_A^{XS} = RT_1 \ln \gamma_A(T_1) = RT_2 \ln \gamma_A(T_2)$$

↓

$$\frac{\ln \gamma_A(T_2)}{\ln \gamma_A(T_1)} = \frac{T_1}{T_2}$$

$$\bar{H}_B = \Delta \bar{H}^M + (1-x_B) \frac{d\Delta \bar{H}^M}{dx_B} \xrightarrow{\Delta \bar{H}^M = \alpha x_A x_B} \bar{H}_B = \alpha x_A^2$$

$$x_i \Delta \bar{S}_i^M = -R \ln x_i$$

$$RT \Delta \bar{G}_A^M = RT \ln a_A = \bar{H}_A^M - T \Delta \bar{S}_A^M \Rightarrow RT \ln a_A = \bar{H}_A^M = \alpha x_B^2$$

$$\begin{matrix} \swarrow & \searrow \\ RT \ln \gamma_A & RT \ln x_A \end{matrix} \quad \begin{matrix} \swarrow \\ RT \ln x_A \end{matrix}$$

$$\Rightarrow RT \ln \gamma_A = \alpha x_B^2 \Rightarrow \gamma_A = \exp\left(\frac{\alpha x_B^2}{RT}\right) \quad \gamma_B = \exp\left(\frac{\alpha x_A^2}{RT}\right)$$

As $x_A \rightarrow 1$, $\gamma_A = \exp\left(\frac{\alpha \cdot 0}{RT}\right) \rightarrow 1$, A is a Raoultian solution

As $x_A \rightarrow 0$, $\gamma_A = \exp\left(\frac{\alpha \cdot 1}{RT}\right) \rightarrow a$ constant, A is a Henrian solution

As $x_B \rightarrow 1$, $\gamma_B = \exp\left(\frac{\alpha \cdot 0}{RT}\right) \rightarrow 1$, B is a Raoultian solution

As $x_B \rightarrow 0$, $\gamma_B = \exp\left(\frac{\alpha \cdot 1}{RT}\right) \rightarrow a$ constant, B is a Henrian solution

9.10 Statistical Model of Solutions: ~~假设~~ $\Delta \bar{V}_i^M = 0$

$$E = P_{AA} E_{AA} + P_{AB} E_{AB} + P_{BB} E_{BB} \quad ZNA = P_{AB} + ZPA_{AA}$$

Δ 数量 Δ 数量
 Δ 数量 Δ 数量

$$\Rightarrow E = \frac{1}{2} ZNA E_{AA} + \frac{1}{2} ZNB E_{BB} + P_{AB} [E_{AB} - \frac{1}{2} (E_{AA} + E_{BB})]$$

before mixing: $ZPA_{AA} = ZNA$ $ZPBB = ZNB$

ΔE^M = the energy of solution - the energy of unmixed components

$$= P_{AB} [E_{AB} - \frac{1}{2} (E_{AA} + E_{BB})] \quad (P_{AB} = ZPA_{AA})$$

for mixing process: $\Delta H^M = \Delta E^M - P \Delta V^M$ (由于 $\Delta V^M = 0$)

$$\Delta H^M = P_{AB} [E_{AB} - \frac{1}{2} (E_{AA} + E_{BB})]$$

$\downarrow$ ideal $\Delta H^M = 0$

$$E_{AB} = \frac{1}{2} (E_{AA} + E_{BB})$$

$\therefore$ if $|E_{AB}| > \frac{1}{2} |E_{AA} + E_{BB}|$ ΔH^M 是负的 eg: Fe-Ni ✓

if $|E_{AB}| < \frac{1}{2} |E_{AA} + E_{BB}|$ ΔH^M 是正的 eg: Fe-Cu ✓

if $\Delta H^M = 0$ 则 A 和 B 随机混合. $\Delta S^M = \Delta S^{M, id} = -R (X_A \ln X_A + X_B \ln X_B)$

$$X_A^2 + 2X_A X_B + X_B^2 = 1 \quad \text{找键几率问题, 可看 PPT}$$

$\downarrow$

$$P_{AB} = \frac{1}{2} ZN_0 \times 2X_A X_B = ZN_0 X_A X_B$$

$$P_{AA} = \frac{1}{2} ZN_0 X_A^2$$

$$P_{BB} = \frac{1}{2} ZN_0 X_B^2$$

$$\therefore \Delta H^M = ZN_0 X_A X_B [E_{AB} - \frac{1}{2} (E_{AA} + E_{BB})]$$

W.

由于是随机混合：所以同 regular solution model: $\Delta H^M = G^XS = z N_0 W_1 X_A X_B$

$$\Delta H^M \propto X_A X_B$$

$$\alpha = z N_0 W_1$$

9.11 subregular Solutions:

$$\Delta H^M = \alpha X_A X_B$$

$$\Delta H^M = z N_0 W_1 X_A X_B$$

$$\alpha = a + b X_B + c X_B^2 + d X_B^3 + \dots$$

$$\downarrow G^XS \text{ 可写为 } G^XS = (a + b X_B) X_A X_B \quad \begin{array}{l} \bar{G}_A = a - X_B \frac{dG}{dX_B} \\ \bar{G}_B = a + X_A \frac{dG}{dX_B} \end{array} \rightarrow \begin{array}{l} \bar{G}_A^{XS} = a X_B^2 + b X_B^2 (X_B - X_A) \\ \bar{G}_B^{XS} = a X_A^2 + 2b X_A^2 X_B \end{array}$$

$$\text{当 } \frac{dG^XS}{dX_B} = 0 \text{ 时曲线出现极值 即 } X_B = \frac{z(b-a) \pm z \sqrt{b^2 + ab + a^2}}{bb} \text{ 时}$$

$\downarrow$ 引入常数 τ

$$G^XS = (a_0 + b_0 X_B) X_A X_B (1 - \frac{\tau}{T})$$

$$S^{XS} = - \frac{\partial G^XS}{\partial T} = \frac{(a_0 + b_0) X_A X_B}{T}$$

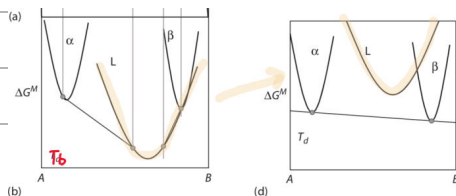
$$\Delta H^M = G^XS + T S^{XS} = (a_0 + b_0 X_B) X_A X_B (2 - \frac{T}{T_d})$$

Chapter 10. 纯. 非混合 component 的 $G=0$

多相平衡时 G_i 都相同

※ 图表: $L \rightarrow (T \downarrow)$

且曲线液相比固相 ΔG^M 上升得快



$$G_A(s) - G_A(l) = -\Delta G_m^{\circ}(A) = -(\Delta H_m^{\circ}(A) - T\Delta S_m^{\circ}(A)) \quad \text{melting: } s \rightarrow l$$

$$\Delta G = \Delta G_l - \Delta G_s$$

$$\Delta G_m^{\circ}(A) = \Delta H_m^{\circ}(A) \left(\frac{T_m(A) - T}{T_m(A)} \right) \quad \text{开始 melt 时 温度}$$

$$\text{前提: } C_p(s) = C_p(l)$$

形成一个均匀的 liquid solution from pure liquid A 和 pure solid B 需 2 步:

$$\textcircled{1} \text{ B 的 melting: } \Delta G = X_B \Delta G_m^{\circ}(B) \quad \text{其中 } \Delta G_m^{\circ}(B) = G_{B(l)}^{\circ} - G_{B(s)}^{\circ} = \Delta H_m^{\circ}(B) \left(\frac{T_m - T}{T_m} \right)$$

$$\textcircled{2} \text{ liquid B 和 liquid A 混合: } \Delta G = \Delta G^{\text{mid}} = RT(X_A \ln X_A + X_B \ln X_B)$$

$$\therefore \Delta G_{\text{mix}} = \Delta G_m^{\circ}(B) = RT(X_A \ln X_A + X_B \ln X_B) + X_B \Delta G_m^{\circ}(B)$$

AB 形成 ideal solid: (A 和 B 的)

$$\Delta G_m^{\circ}(s) = RT(X_A \ln X_A + X_B \ln X_B) - X_A \Delta G_m^{\circ}(A)$$

$$\star \bar{G}_B = G_{B(l)}^{\circ} + RT \ln(a_B) \quad \rightarrow \text{with respect to liquid B}$$

$$\bar{G}_B = G_{B(s)}^{\circ} + RT \ln(a_B) \quad \rightarrow \text{with respect to solid B}$$

$$\Delta G_m^{\circ}(B) = G_{B(l)}^{\circ} - G_{B(s)}^{\circ} = RT \ln \left(\frac{a_{B(l)}}{a_{B(s)}} \right)$$

$$\frac{a_{l(i)}}{a_{s(i)}} = \exp \left(\frac{\Delta G_m^{\circ}(i)}{RT} \right) = \exp \left(\frac{\Delta H_m^{\circ}(i)}{RT} \left(\frac{T_m(i) - T}{T_m(i)} \right) \right) \Rightarrow \ln a_i = \frac{-\Delta H_m^{\circ}(i)}{RT} + \frac{\Delta H_m^{\circ}(i)}{RT_m(i)}$$

$$\Delta G^{\text{M}} - \Delta G^{\text{mid}} = G^{\text{XS}} = \Delta H^{\text{M}} = \alpha X_A X_B$$

$$\Delta G^{\text{M}} = RT(X_A \ln X_A + X_B \ln X_B) + \alpha X_A X_B$$

$$\left. \begin{aligned} \frac{\partial \Delta G^{\text{M}}}{\partial X_B} &= 0 \\ \frac{\partial \Delta G^{\text{M}}}{\partial X_A} &= 0 \end{aligned} \right\} \Rightarrow \frac{\alpha}{RT} = 2$$

T_{cr}

summary:

$$\textcircled{1} \Delta G^{\text{M}} = RT(X_A \ln a_A + X_B \ln a_B) \xrightarrow{\text{ideal}} \Delta G^{\text{M}} = RT(X_A \ln X_A + X_B \ln X_B)$$

② for regular solution: $\Delta G^M - \Delta G^M \cdot \text{id} = G^XS = \alpha X_A X_B = \Delta H^M$

③ 二元体系A-B中判断 α 相 β 相平衡的依据: $a_A(\text{in } \alpha) = a_A(\text{in } \beta)$
 $a_B(\text{in } \alpha) = a_B(\text{in } \beta)$

④ 形成 ideal liquid solutions 和 ideal solid solutions, 在 A-B 二元相图中:

固相线: $X_{A(s)} = \frac{1 - \exp(-\frac{\Delta G_m^{(B)}}{RT})}{\exp(-\frac{\Delta G_m^{(A)}}{RT}) - \exp(-\frac{\Delta G_m^{(B)}}{RT})}$

液相线:

$$X_{A(l)} = \frac{[1 - \exp(-\frac{\Delta G_m^{(B)}}{RT})] \exp(-\frac{\Delta G_m^{(A)}}{RT})}{\exp(-\frac{\Delta G_m^{(A)}}{RT}) - \exp(-\frac{\Delta G_m^{(B)}}{RT})}$$

$$X_{A(l)} = X_{A(s)} \cdot \exp(-\frac{\Delta G_m^{(A)}}{RT})$$

⑤ 如果 liquid solution ideal, $-\Delta G_m^{(A)} = RT \ln X_A$

regular, $-\Delta G_m^{(A)} = RT \ln X_A + \alpha(1-X_A)^2$

chapter 11 涉及气体的反应.



AB 2 气体混合, 混合前 $\bar{G}_i^{\text{mix}} = RT \ln X_i$, 则 $\bar{G}_i = \bar{G}_i^\circ + RT \ln P + RT \ln X_i$

假设 1 mol A 1 mol B 为初始. 反应后 在总体系中的占比

乘 $n_A = n_B$ 则 $n_C = 2 - n_A - n_B$ (生成)

且 $X_A = \frac{n_A}{2 - n_A} \quad X_B = \frac{n_B}{2 - n_A} \quad n_C = \frac{2(1-n_A)}{2 - n_A}$

联立

$$G' = n_A \bar{G}_A + n_B \bar{G}_B + 2(1-n_A) \bar{G}_C$$

$$G' = n_A (\bar{G}_A^\circ - RT \ln P + RT \ln X_A) + n_B (\bar{G}_B^\circ - RT \ln P + RT \ln X_B) + 2(1-n_A) (\bar{G}_C^\circ - RT \ln P + RT \ln X_C)$$

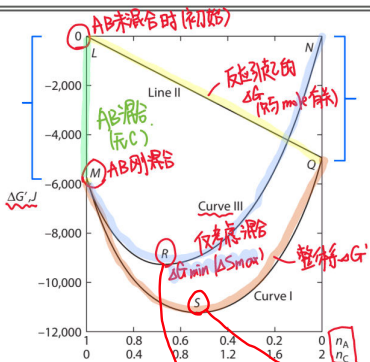
反应物
生成物
A+B
nA
2x5

$$G' - 2G_C^\circ = n_A (-\Delta G^\circ) + 2RT \ln P$$

$$+ 2RT \left[n_A \ln \left(\frac{n_A}{2} \right) + (1-n_A) \ln (1-n_A) \right]$$

$$\Delta G^\circ = 2\Delta G_C^\circ - \Delta G_A^\circ - \Delta G_B^\circ$$

ΔG : ① AB混合 G_L ② 化学反应产生 ΔG



$$\Delta G' (L-M) = RT \left[n_A \ln \left(\frac{P_A}{P_A^*} \right) + n_B \ln \left(\frac{P_B}{P_B^*} \right) \right]$$

仅AB混合
不反应

mix 中的
pure

$$n_A = n_B = 1, \quad P_A = P_B = P_{\text{mixture}} = 1$$

$$\therefore P_A = P_B = \frac{1}{2}$$

$$\Delta G' (L-M) = kT \times 2 \times \ln 0.5$$

$$\frac{dG'}{dn_A} = 0 \quad (\text{看 mix 和 反应})$$

焓最大 (纯 mix)

$$\text{且 } G' = n_A \bar{G}_A + n_B \bar{G}_B + 2(1-n_A) \bar{G}_C$$

$$\frac{\partial G'}{\partial n_A} = \bar{G}_A + \bar{G}_B - 2\bar{G}_C = 0 \quad \therefore \bar{G}_A + \bar{G}_B = 2\bar{G}_C \text{ 时平衡}$$

$$\therefore (G_A^0 + RT \ln P_A) + (G_B^0 + RT \ln P_B) = 2G_C^0 + 2RT \ln P_C$$

$$2G_C^0 - G_B^0 - G_A^0 = \Delta G^0 = -RT \ln \left(\frac{P_C^2}{P_A P_B} \right) \quad k_p = \frac{P_C^2}{P_A P_B} \quad \therefore \Delta G^0 = -RT \ln k_p$$

$$\left. \begin{aligned} LM &= 2RT \ln 0.5 \\ NQ &= \Delta G^0 = f(T) \end{aligned} \right\}$$

$$NQ = \Delta G^0 = f(T)$$

$$\left[\frac{\partial \Delta G^0}{\partial T} \right]_P = -\Delta S^0 \quad \left[\frac{\partial \Delta G^0}{\partial T} \right]_P = -\frac{\Delta H^0}{T^2} \quad \Delta G^0 = -RT \ln k_p \quad \frac{d \ln k_p}{d \left(\frac{1}{T} \right)} = \frac{-\Delta H^0}{R}$$

$\Delta H^0 > 0$ 时 $T \uparrow \rightarrow k_p \uparrow$ 吸热; $\Delta H^0 < 0$ $T \uparrow \rightarrow k_p \downarrow$ 放热.

$$\text{eg. } 2\text{Cl} = \text{Cl}_2 \quad k_p = \frac{P_{\text{Cl}_2}}{P_{\text{Cl}}^2}$$

$$\text{Cl}_2 = 2\text{Cl} \quad k_p = \frac{P_{\text{Cl}}^2}{P_{\text{Cl}_2}} = \frac{\chi_{\text{Cl}}^2 \cdot P^2}{\chi_{\text{Cl}_2} P} = k_x P$$

$$A + B = 2C \quad k_p = \frac{P_C^2}{P_A P_B} = \frac{\chi_C^2 P^2}{(\chi_A P)(\chi_B P)} = \frac{\chi_C^2}{\chi_A \chi_B} = k_x$$

$$aA + bB = cC + dD \quad k_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} = \frac{X_C^c X_D^d}{X_A^a X_B^b} P^{(c+d-a-b)} \quad k_x$$

$$A + B = 2C: \quad G' - 2G_C^\circ = n_A(-41^\circ) + n_A(T \cdot \Delta S^\circ) + 2RT \left[n_A \left(\ln \frac{n_A}{2} \right) + (1+n_A) \ln(1+n_A) \right] \quad \text{mixing}$$

例: 反应 $SO_2(g) + \frac{1}{2}O_2(g) = SO_3(g)$, $\Delta G^\circ = -74600 + 89.3 T$ J

$$T = 1000 \text{ K}, \Delta G_{1000}^\circ = -5250 \text{ J}, \ln k_p = \frac{-5250}{8.314 \times 1000} = -0.629$$

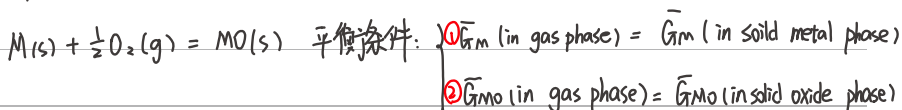
$$k_p = 1.876 = \frac{P_{SO_3}}{P_{SO_2} P_{O_2}^{1/2}}; \quad \begin{array}{ccc} SO_2 & + \frac{1}{2} O_2 & = SO_3 \\ 1 & \frac{1}{2} & 0 \\ 1-x & \frac{1}{2} \cdot \frac{1}{2}x & x \end{array} \quad n_{tot} = 1-x + \left(\frac{1}{2} - \frac{1}{2}x\right) + x = \frac{1}{2}(3-x)$$

$$\text{且 } P_i = \frac{n_i}{n_T} \cdot P, \text{ so } k_p = \frac{P_{SO_3}}{P_{SO_2}^2 P_{O_2}} = \frac{(3-x)x^2}{(1-x)^3 P}$$

$$k_p = \exp\left(-\frac{\Delta G^\circ}{RT}\right)$$

例: a cm³ 的 A 在 1 atm 和 b cm³ 的 B 在 1 atm 混合: $P_A = \frac{n_A RT}{(a+b)} \Rightarrow \frac{P_A}{P_B} = \frac{n_A}{n_B} = \frac{a}{b}$

Chapter 12:



$$\text{且 } \Delta G^\circ = G_{MO(g)}^\circ - \frac{1}{2}G_{O_2(g)}^\circ - G_{M(g)}^\circ = -RT \ln \frac{P_{MO}}{P_M \cdot P_{O_2}^{1/2}}$$

$$\textcircled{1} \text{式: } G_M^\circ + RT \ln P_{M(g)} = G_M(s) + \int_{P=1}^{P=P_{M(g)}} V_{M(s)} dp \quad \text{可忽略}$$

$$\textcircled{2} \text{式: } G_{MO(g)}^\circ + RT \ln P_{MO(g)} = G_{MO(s)} + \int_{P=1}^{P=P_{MO(g)}} V_{MO} dp$$

$$\text{该反应 } \Delta G^\circ = -RT \ln \left(\frac{1}{P_{O_2}^{1/2}} \right) = -RT \ln k$$

$$F = C + 2 - \Phi$$

$\downarrow$ $\downarrow$
 相和 相数
 > 1

$$\Delta G_T^\circ = \Delta H_T^\circ - T \Delta S_T^\circ = \Delta H_{298}^\circ + \int_{298}^T \Delta C_p dT - T \Delta S_{298}^\circ - T \int_{298}^T \frac{\Delta C_p}{T} dT$$

$$\left(\frac{\partial \Delta H^\circ}{\partial T} \right)_p = \Delta C_p = \Delta a + \Delta b T + \Delta c T^2$$

↓ 积分

$$\Delta H_T^\circ = \Delta H_{298}^\circ + \Delta a T + \frac{\Delta b T^2}{2} - \frac{\Delta c}{T}$$

$$\frac{\partial \Delta G^\circ}{\partial T} = -\frac{\Delta H^\circ}{T^2} = -\frac{\Delta H_{298}^\circ}{T^2} - \frac{\Delta a}{T} - \frac{\Delta b}{2} + \frac{\Delta c}{T^3} \xrightarrow{\text{积分}} \Delta G^\circ = T \Delta H_{298}^\circ - \Delta a T \ln T - \frac{\Delta b T^2}{2} - \frac{\Delta c}{2T}$$

$= -RT \ln K$
 $\Rightarrow \ln K = -\frac{\Delta H_{298}^\circ}{RT} - \frac{1}{K} + \frac{\Delta a \ln T}{R} + \frac{\Delta b T}{2R} + \frac{\Delta c}{2RT^2}$

例: 反应 $A(l) = A(g)$

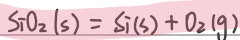
平衡条件: $G_{A(l)}^\circ = G_{A(g)}^\circ + RT \ln P_A$ 则 $\Delta G^\circ = -RT \ln P_A = -RT \ln K$ 平衡

vapor phase 理想: $\Delta G_{\text{evap}} = 0$



$$\Delta G_T^\circ = -RT \ln K = -RT \ln \left(\frac{1}{P_{O_2}(\text{eq. } T)} \right) = \Delta H^\circ - T \Delta S^\circ$$

Chapter B: 含有 condensed solution 的平衡.



$$\bar{G}_{SiO_2} = \bar{G}_{Si} + \bar{G}_{O_2} \quad \text{① } SiO_2 \text{ 和 } Si \text{ 是 pure 的} \rightarrow G_{SiO_2}^\circ = G_{Si}^\circ + G_{O_2}^\circ = G_{Si}^\circ + G_{O_2}^\circ + RT \ln P_{O_2}(\text{eq. } T)$$

② 如果 SiO_2 在 Al_2O_3 - SiO_2 固体中:

$$\bar{a}_{\text{SiO}_2} = \bar{G}_{\text{SiO}_2} + RT \ln a_{\text{SiO}_2}$$

$$\bar{G}_{\text{SiO}_2} + RT \ln a_{\text{SiO}_2} = \bar{G}_{\text{Si}} + \bar{G}_{\text{O}_2} + RT \ln p'_{\text{O}_2}(\text{eq. T})$$

$$p'_{\text{O}_2}(\text{eq. T}) = p_{\text{O}_2}(\text{eq. T}) \cdot a_{\text{SiO}_2}$$

$$aA + bB = cC + dD, \quad \Delta G' = c\bar{G}_C + d\bar{G}_D - a\bar{G}_A - b\bar{G}_B$$

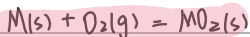
$$\Delta G' - \Delta G^\circ = c(\bar{G}_C - \bar{G}_C^\circ) + d(\bar{G}_D - \bar{G}_D^\circ) - a(\bar{G}_A - \bar{G}_A^\circ) - b(\bar{G}_B - \bar{G}_B^\circ)$$

$$\bar{G}_i = \bar{G}_i^\circ + RT \ln a_i \quad \Delta G' - \Delta G^\circ = c(RT \ln a_C) + d(RT \ln a_D) - a(RT \ln a_A) - b(RT \ln a_B)$$

$$= RT \ln \left(\frac{a_C^c a_D^d}{a_A^a a_B^b} \right) = RT \ln Q \quad Q = \frac{a_C^c a_D^d}{a_A^a a_B^b}$$

$$\Delta G^\circ = -RT \ln k \quad \text{此时 } Q^{\text{eq}} = k$$

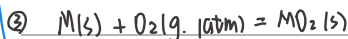
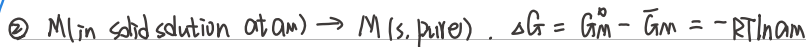
$$a_{\text{O}_2} = \frac{\text{the pressure of oxygen in the gas phase}}{\text{the pressure of oxygen in its standard state}} = 1 \text{ atm}$$



$$Q = \frac{a_{\text{MO}_2}}{a_{\text{M}} a_{\text{O}_2}} \quad \text{① M, MO}_2 \text{ is pure} \rightarrow Q = \frac{1}{a_{\text{O}_2}} = \frac{1}{p_{\text{O}_2}} \quad \text{and } Q^{\text{eq}} = \frac{1}{p_{\text{O}_2}(\text{eq. T})} = k$$

$$\text{② M in solution 中活度为 } a_{\text{M}} \quad Q^{\text{eq}} = \frac{1}{a_{\text{M}} p_{\text{O}_2}(\text{eq. T})} = k$$

$$\text{③ MO}_2 \text{ ————— } a_{\text{MO}_2}, \quad Q^{\text{eq}} = \frac{a_{\text{MO}_2}}{p_{\text{O}_2}(\text{eq. T})} = k$$



$$\Delta G_{\text{rO}} = \Delta G_{\text{②}} + \Delta G_{\text{③}} = \Delta G^\circ - RT \ln a_{\text{M}}$$

$$kT = \frac{1}{p_{\text{O}_2}(\text{T. eq. M, s})} = \frac{1}{a_{\text{M}} p_{\text{O}_2}(\text{T. eq. M (at } a_{\text{M}}) / \text{MO}_2(\text{s}))}$$



$$K_T = \frac{1}{P_{O_2}(T, eq M_{(s)}) / MO_{2(s)}} \\ = \frac{a_{MO_2}}{P_{O_2}(T, eq M_{(s)}) / MO_{2(at a_{MO_2})}}$$

M (in solid solution at a_M) + $O_2(g, 1 atm) = MO_2$ (at a_{MO_2})

$$\Delta G = \Delta G^\circ - RT \ln a_M - RT \ln P_{O_2} + RT \ln a_{MO_2} = \Delta G^\circ + RT \ln Q$$

→ 平衡时 $\Delta G = 0$. 则 $\Delta G^\circ = -RT \ln Q^\circ = -RT \ln K$

[]: metal phase (): oxide phase

当 $x_B \rightarrow 0$ 时, 根据 Henry's 定理, $\frac{a_B}{x_B} \rightarrow \gamma_B^\circ (= k_B)$

→ $h_B = f_B x_B$ (定 x_B)

$$h_B (1 wt\%) = wt\% B$$

① B (in the Raoultian standard state) → B (in the Henrian →)

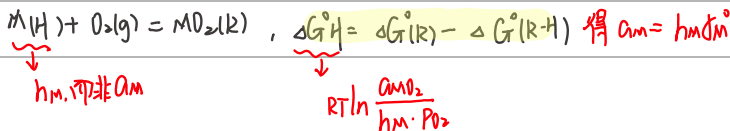
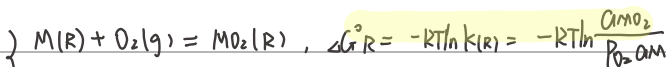
$$\Delta G_B^{\circ}(R-H) = G_B^{\circ}(H) - G_B^{\circ}(R) = RT \ln \frac{a_B(\text{in } H)}{a_B(\text{in } R)} = RT \ln \left(\frac{h_B}{\gamma_B^\circ} \right) = RT \ln \gamma_B^\circ$$

② B (in the Henrian standard state) → B (in the 1 wt% standard state)

$$G_B^{\circ}(H \rightarrow 1 \text{ wt}\%) = G_B^{\circ}(1 \text{ wt}\%) - G_B^{\circ}(H) = RT \ln \frac{a_{B(\text{in the 1 wt\% standard state})}}{a_{B(\text{in the Henrian standard state})}}$$

$$\frac{a_{B(\text{in the 1 wt\% standard state})}}{a_{B(\text{in the Henrian standard state})}} = \frac{w_C}{h_B} = \frac{ac}{ab} = \frac{MW_A}{100 MW_B} \Rightarrow \Delta G_{B(R \rightarrow 1 \text{ wt}\%)}^\circ = RT \ln \left(\frac{\gamma_B^\circ MW_A}{100 MW_B} \right)$$

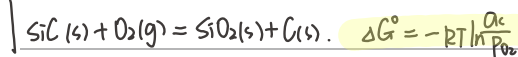
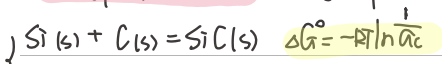
solvent



$$F = C + 2 - \Phi$$

$\downarrow$ component $\rightarrow$ phase

the equilibrium Si-SiC-gas phase:



$$2.303 \log(x) = \ln(x)$$

In the field of stability of Si: $\text{Si}(s) + \text{CO}_2(g) = \text{SiO}(g) + \text{CO}(g)$

$$\Delta G^\circ = -RT \times 2.303 \log \frac{P_{\text{SiO}} P_{\text{CO}}}{P_{\text{CO}_2}}$$

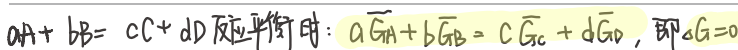
In the field of stability of SiO₂: $\text{SiO}_2(s) + \text{CO}(g) = \text{SiO}(g) + \text{CO}_2(g)$

$$\Delta G^\circ = -RT \times 2.303 \log \frac{P_{\text{CO}_2} P_{\text{SiO}}}{P_{\text{CO}}}$$



ln

$\rightarrow$ 浓度单位: wt%



$$\Delta G^\circ = -RT \ln K = -RT \ln \frac{a_C^c a_D^d}{a_A^a a_B^b}$$

Raoultian standard: $X_B \rightarrow 0$, $\frac{a_B}{X_B} \rightarrow k_B / \gamma_B^\circ$

Henrian standard: $X_B \rightarrow 1$, $a_B = \gamma_B^\circ X_B = \bar{\gamma}_B^\circ$, $h_B = f_B X_B$

wt%-standard: $\frac{h_B(1 \text{ wt}\%)}{\text{wt}\% B} \rightarrow 1$ as $\text{wt}\% B \rightarrow 0$, $h_B(1 \text{ wt}\%) = f_B(1 \text{ wt}\%) \text{ wt}\% B$, $a_B = h_B \bar{\gamma}_B^\circ$

$$a_B = f_B(1 \text{ wt}\%) \cdot \text{wt}\% B \cdot \bar{\gamma}_B^\circ \cdot \frac{\text{MW}_{\text{solvent}}}{100 \text{ MW}_B}$$

$$F = (N - R) + 2 - \Phi, \quad C = N - R$$

$$\ln f_B = \varepsilon_B^B X_B + \varepsilon_B^C X_C + \varepsilon_B^D X_D \quad \text{其中} \quad \varepsilon_i^j = \left. \frac{\partial \ln f_j}{\partial X_i} \right|_{X_i \rightarrow 0}$$

★ Raoultian standard: 溶剂在极其稳定下的纯净物质

$$p_A = p_A^* \cdot x_A, \quad a_A = x_A$$

理想
溶液中
A 蒸气压
A 蒸气压

★ Henry law: 无限稀薄的溶液中, 溶剂活度 $\propto$ 溶剂摩尔质量

$$p_B = k_B^{(B)} x_B, \quad a_B = k_B \cdot x_B, \quad \frac{a_B}{x_B} \rightarrow f_B^{\circ} (x_B \rightarrow 0)$$

B 分压 Henry 常数 活度系数

$x_B \rightarrow 1 \rightarrow$ 得到 Henrian standard, 此时 $a_B = f_B^{\circ}$, $h_B = f_B^{\circ} x_B$

★ 活度定义: $a_i = f_i \cdot x_i$, 理想溶液, 纯溶剂则 $a_i = 1$

